TASK2 CSE281

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part1 :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("task2.jpg", cv2.IMREAD_GRAYSCALE)
if img is None:
    print("Image not found!")
else:
    dark = (img * 0.4).astype(np.uint8)
    bright = np.clip(img * 1.5, 0, 255).astype(np.uint8)
    low_contrast = np.clip(img * 0.3 + 100, 0, 255).astype(np.uint8)

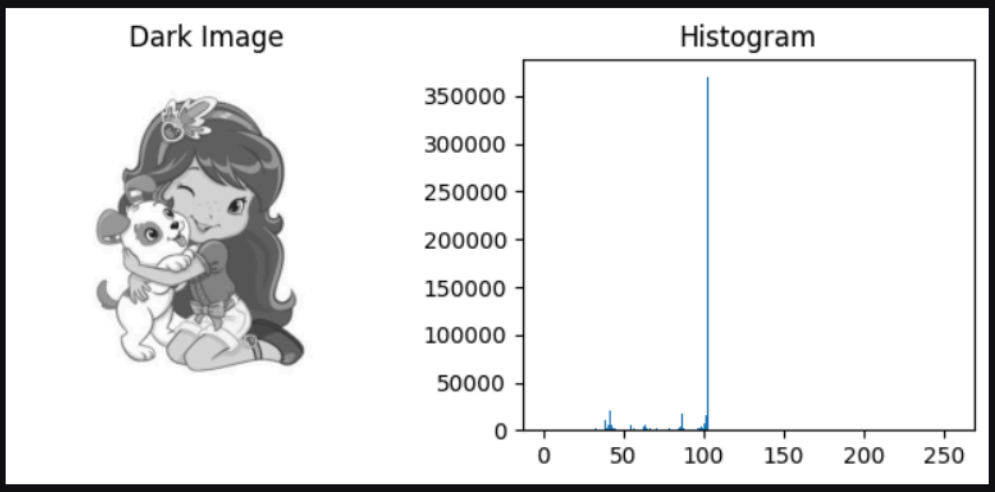
    images = [dark, bright, low_contrast]
    names = ["Dark Image", "Bright Image", "Low Contrast Image"]
    for i in range(3):
        plt.figure(figsize=(8, 3))

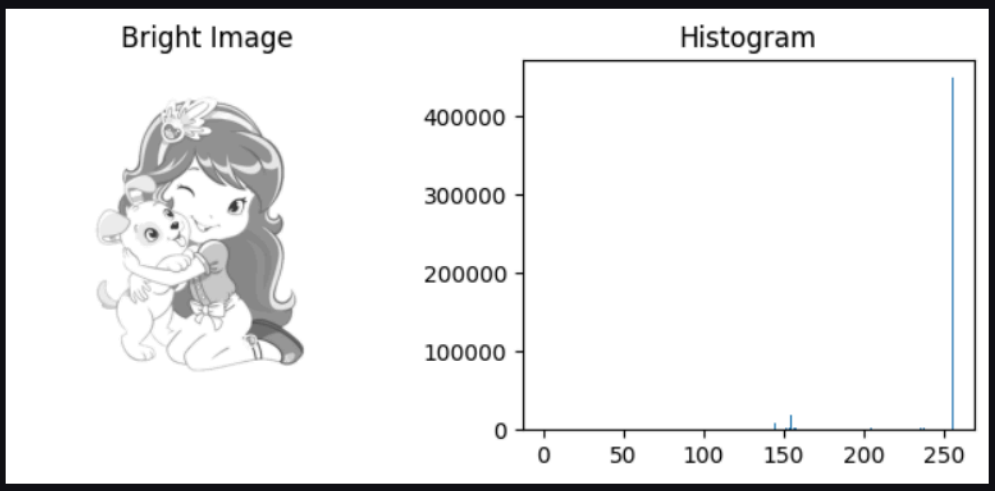
        plt.subplot(1, 2, 1)
        plt.imshow(images[i], cmap='gray')
        plt.title(names[i])
        plt.axis('off')

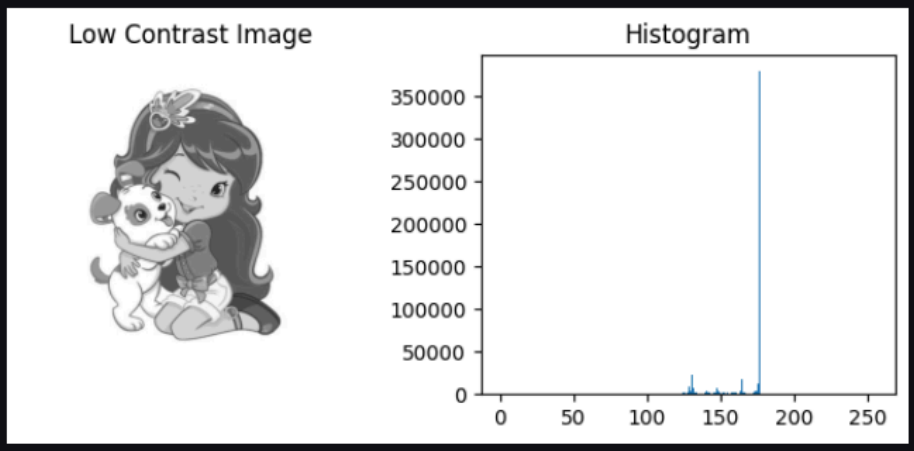
        plt.subplot(1, 2, 2)
        plt.hist(images[i].ravel(), bins=256, range=(0, 256))
        plt.title("Histogram")

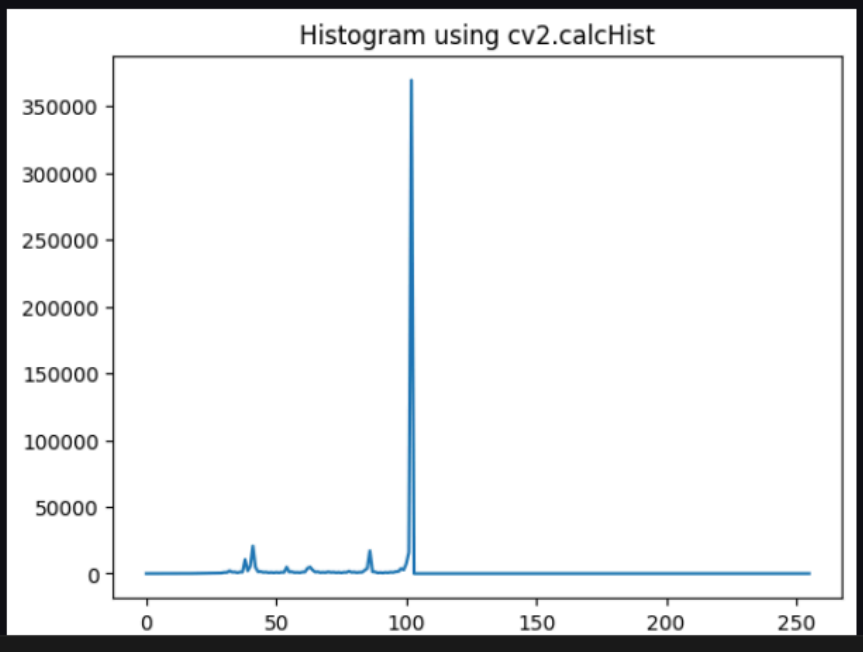
        plt.show()
    hist = cv2.calcHist([dark], [0], None, [256], [0, 256])
    plt.plot(hist)
    plt.title("Histogram using cv2.calcHist")
    plt.show()
```

✓ 12.0s


Dark Image
Histogram
350000
300000
250000
200000
150000
100000
50000
0
0
50
100
150
200
250


Bright Image
Histogram
400000
300000
200000
100000
0
0
50
100
150
200
250



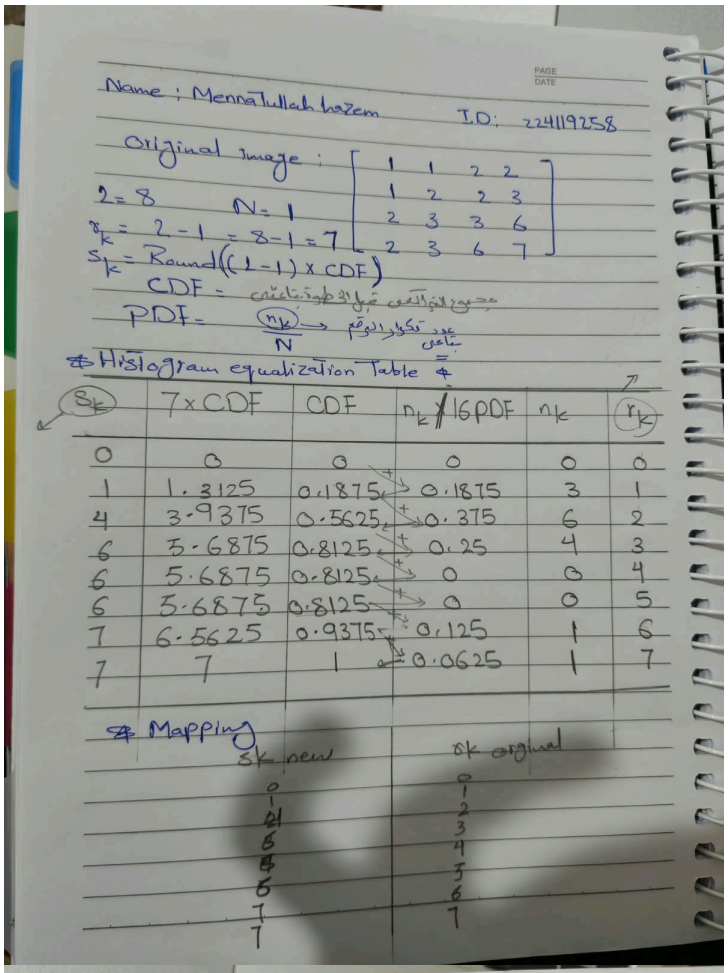


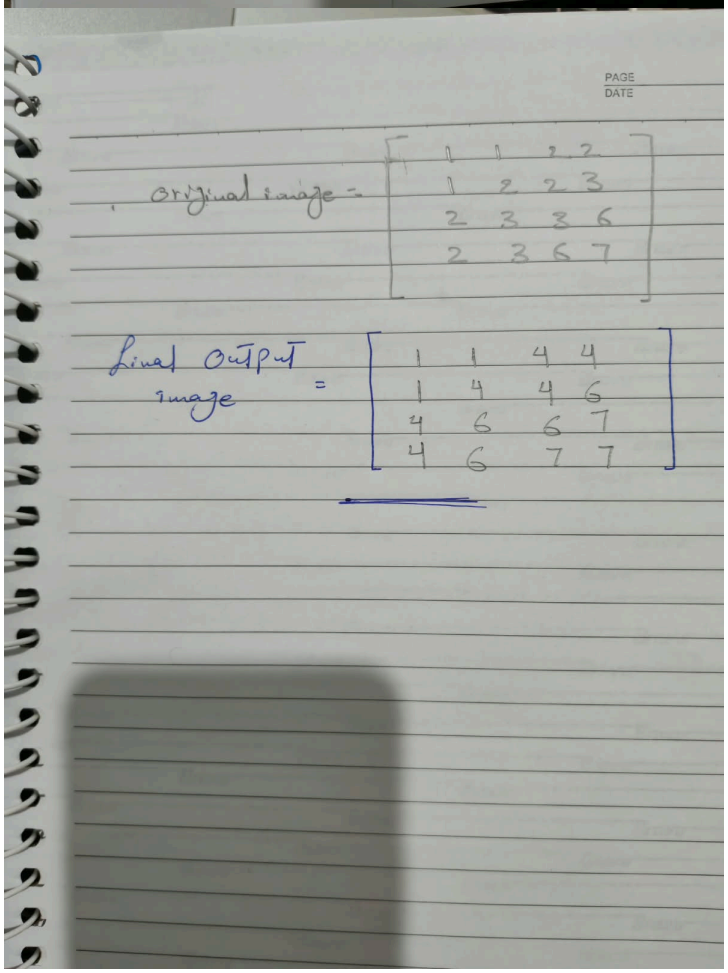

```
dark => most pixels concentrated on left side  bec the low intensity values
bright => most pixels concentrated on right side  bec the high intensity values
low contrast => pixels concentrated in a narrow range bec the small diffrence between pixels intensites
```

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part 2:

on paper:


Name : Mennatullah hazem
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Original image :
1 1 2 2
1 2 2 3
2 3 3 6
2 3 6 7
L = 8 N = 1
$r_k = 2 - 1 = 8 - 1 = 7$
S_k = Round((L-1) x CDF)
CDF =
PDF = $\frac{n_k}{N}$
Histogram equalization Table
Sk | 7xCDF | CDF | nk/16PDF | nk | rk
0 | 0 | 0 | 0 | 0 | 0
1 | 1.3125 | 0.1875 | 0.1875 | 3 | 1
4 | 3.9375 | 0.5625 | 0.375 | 6 | 2
6 | 5.6875 | 0.8125 | 0.25 | 4 | 3
6 | 5.6875 | 0.8125 | 0 | 0 | 4
6 | 5.6875 | 0.8125 | 0 | 0 | 5
7 | 6.5625 | 0.9375 | 0.125 | 1 | 6
7 | 7 | 1 | 0.0625 | 1 | 7
Mapping
Sk new
0
1
4
6
6
6
7
7
Sk original
0
1
2
3
4
5
6
7


original image =
1 1 2 2
1 2 2 3
2 3 3 6
2 3 6 7
Final Output image =
1 1 4 4
1 4 4 6
4 6 6 7
4 6 7 7

CODE:

```python
import numpy as np

def my_equalize(img, L=8):

    N = img.size
    hist = np.zeros(L)

    for pixel in img.ravel():
        hist[pixel] += 1
    pdf = hist / N
    cdf = np.cumsum(pdf)
    sk = np.round((L - 1) * cdf).astype(int)

    output = sk[img]

    return output

img = np.array([
    [1, 1, 2, 2],
    [1, 2, 2, 3],
    [2, 3, 3, 6],
    [2, 3, 6, 7]
])
result = my_equalize(img, L=8)

print("Original Image:")
print(img)

print("Equalized Image:")
print(result)
```

✓ 0.0s

```
Original Image:
[[1 1 2 2]
 [1 2 2 3]
 [2 3 3 6]
 [2 3 6 7]]
Equalized Image:
[[1 1 4 4]
 [1 4 4 6]
 [4 6 6 7]
 [4 6 7 7]]
```